Yuni Wu

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Summary

Machine Learning Engineer focused on dependable model behavior across computer vision, robot learning, and inference systems. Builds evaluation workflows that preserve measurable quality under class imbalance, latency constraints, deployment validation, and production reliability requirements.

Experience

Graduate Research Assistant Sept 2025 – May 2026
CAIRO Lab, University of Colorado Boulder — Robotics and embodied AI research lab

- Improved autonomous waypoint prediction to 0.11 MAE and $R^2 = 0.62$ by fine-tuning PyTorch/torchvision ResNet-18 behavior-cloning policies on 97K+ robot state-action samples.
- Converted ROS/rosbag telemetry into reproducible robot-learning datasets by aligning camera imagery, motion-command supervision, checkpoints, k-fold validation, and model-selection metrics.
- Validated Unitree Go1 inference behavior with ensembling, test-time augmentation, conservative smoothing, latency checks, stability validation, and safety monitoring.

Machine Learning Engineer Intern May 2025 – Aug 2025
MyEdMaster LLC — AI-driven healthcare and education technology company

- Verified 0.92 F1-score on a 22-class medical-image classification workflow by evaluating ResNet-based models across 50K+ imbalanced samples with repeatable validation scripts.
- Reduced per-image inference latency by 5× (0.15s to 0.03s) by comparing ResNet architectures and selecting faster model variants without dropping reliability checks.
- Diagnosed model reliability with cross-validation, ROC-AUC, calibration, confusion-matrix analysis, and regression-style evaluation workflows.

Software Engineer June 2021 – Aug 2022
Shopee Co. Ltd. — Major Southeast Asian e-commerce platform

- Reduced manual QA workload by 80% by engineering Go API validation, SQL checks, and anomaly-detection workflows across 800+ production marketplace scenarios.
- Accelerated reporting speed by 78% by building regression monitoring pipelines that surfaced data inconsistencies earlier.

Projects

NailAI — Cloud-Native ML Inference Platform [demo]

- Powered asynchronous image inference with FastAPI-backed Cloud Run services, Pub/Sub job routing, Cloud Storage asset handling, and BigQuery prediction metadata.
- Recorded 0 failures in a load test at 200 concurrent users with p95 latency of 6.8s through Pub/Sub job routing, retry handling, and backend/worker coordination.
- Instrumented request, job, and prediction metadata for inference monitoring, downstream analysis, and reliability investigation.

Multimodal Vision-Language Navigation for Unitree Go1 [project]

- Benchmarked Qwen3-VL-30B and InternVL-3.5-14B across 13 Go1 social-navigation rosbags and 2.4K+ labeled state-action sequences.
- Measured uncertainty handling, directional activation, crossing detection, unsafe-forward behavior, latency, and bag-level consensus accuracy.

Technical Skills

Languages: Python, SQL, Go, R, TypeScript
ML: PyTorch, torchvision, NumPy, ResNet/CNNs, behavior cloning, transfer learning, class-imbalance handling, hierarchical classification
Evaluation: cross-validation, ROC-AUC, F1-score, calibration, confusion matrices, benchmarking, regression validation, latency analysis
Systems: FastAPI, Docker, Cloud Run, Pub/Sub, Cloud Storage, BigQuery, metadata pipelines, monitoring, logging, LangGraph, RAG, Kubernetes

Education

University of Colorado Boulder — MS, Data Science | GPA: 3.76/4.0 May 2026
Soochow University — BS, Data Science | GPA: 3.7/4.0 June 2021